

# Brandon Fluegel, PhD

STAFF HUMAN FACTORS ENGINEER | HUMAN-AGENT INTERACTION + COGNITIVE SYSTEMS

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[www.humanfactors.pro](http://www.humanfactors.pro)

Human Factors Engineer and Researcher (Ph.D.) specializing in Human Systems Integration, Physical AI, and safety-critical system validation. Over 6 years of industry experience applying psychophysics, neuroimaging, and cognitive load modeling to bridge the gap between human capabilities and autonomous system performance at **NASA**, **Mercedes-Benz**, **Amazon**, and **Sling**. Passionate about making complex non-deterministic systems safe, predictable, and simple to use.

## PROFESSIONAL EXPERIENCE

### Sling

Jun 2025 to Present

#### STAFF HUMAN FACTORS ENGINEER & RESEARCHER

- Lead organizational human factors strategy and design validation across core software, physical hardware, and AI-driven platforms, directly translating perceptual science into engineering requirements.
- Developed the "Principles for Agentic Trust" framework—accepted for presentation and publication at the **ACM CSCW 2026** Industry Perspectives track—to audit non-deterministic AI systems across four phases (Alignment, Execution, Control, Calibration). This framework defines actionable metrics for system observability, human-in-the-loop (HITL) feedback, predictability, and safe manual overrides.
- Direct physical ergonomics, reach envelope modeling, and multimodal usability protocols across the hardware portfolio. Identify physical use-errors and define anatomical safety, interaction architecture, and mechanical fit specifications to ensure intuitive and safe physical interactions.
- Established robust usability criteria for system latency, auditory/visual feedback, and spatial layouts, grounding engineering performance targets in human perceptual thresholds to minimize cognitive friction.

### Amazon, Devices Design Group

Jun 2020 to Nov 2024

#### HUMAN FACTORS DESIGN ENGINEER & RESEARCHER

- Awarded **US Patent (US-12532040-B1)** and a 2023 Amazon Inventor Award for novel context-aware multimodal interaction architectures—optimizing physical inputs based on dynamic system states.
- Led a multi-year psychophysics program to replace arbitrary engineering latency targets with human perception-derived thresholds. Defined exactly when system delay degrades human trust, influencing hardware responsiveness criteria for multimodal interactions and projecting \$50M in operational value.
- Co-established an objective framework measuring cognitive load in complex UIs using fNIRS neuroimaging and eye-tracking. Quantified neural activation to identify operational workload limits and refine interaction pacing.
- Directly shaped product and interaction strategy for the Echo Show and Echo Hub physical product lines (reaching 75M+ customers). Championed human-centered design principles to architect expressive, communicative behaviors for hardware.

### NASA Langley Research Center

Jun 2018 to Jan 2019

#### HUMAN FACTORS PH.D. INTERN (SYSTEMS SAFETY & WORKSTATIONS)

- Led safety-critical Human-Systems Integration (HSI) and human factors design validation for next-generation clinical workstations on the Lunar Gateway, simulating microgravity operational environments.
- Conducted detailed Usability Failure Modes and Effects Analysis (uFMEA) to identify, trace, and mitigate high-consequence use errors, spatial layout risks, and cognitive load bottlenecks in high-stakes environments.
- Redesigned physical layouts and control displays per NASA-STD-3001 and MIL-STD-1472 guidelines, reducing operator task completion times by 30% while eliminating critical system input errors.

## ADDITIONAL RESEARCH & SYSTEMS EXPERIENCE

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### Mercedes-Benz Research & Development

Jun 2017 to Aug 2017

#### HUMAN FACTORS PH.D. INTERN (AUTONOMOUS SYSTEMS HANDOVERS)

- Designed and executed high-fidelity simulator studies utilizing biometric, auditory, and tactile/haptic alerts to assess human situational awareness, predictability, and reaction times during L2/L3 autonomous vehicle control handovers.
- Developed system-level interaction guidelines for non-deterministic behavior that optimized driver response curves and increased control-transition safety and trust ratings by 24%.

### Uber

Jun 2019 to Sep 2019

#### HUMAN FACTORS PH.D. INTERN (MOBILE HMI & NAVIGATION)

- Conducted naturalistic, real-world field studies and usability evaluations in high-stress urban environments to assess complex spatial navigation and driver-partner HMI.
- Redesigned core information-architecture workflows to decrease cognitive load and visual distraction during dynamic tasks, driving a 5% increase in driver retention metrics.

### Brigham & Women's Hospital / Harvard Medical School

Jun 2014 to Jul 2015

#### TECHNICAL RESEARCH ASSISTANT: CLINICAL NEUROSCIENCE LABORATORY

- Managed large-scale functional neuroimaging (fMRI) workflows and the processing of multi-channel bio-behavioral data, including ECG and cognitive performance telemetry, to measure human responses to acute stress.
- Ensured strict compliance with federal clinical research standards and IRB protocols when managing complex longitudinal data for large patient cohorts.

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## EDUCATION

### Old Dominion University

#### Ph.D. in Human Factors Psychology (2020)

Dissertation quantified the limits of human working memory, spatial processing, and reaction times during high-stress, safety-critical multimodal task interruptions.

#### M.S. in Human Factors Psychology (2017)

## PATENTS & AWARDS

- **Amazon Inventor Award & Patent US-12532040-B1 (2023):** Recognized for developing novel multimodal interaction architectures for context-based user interface control inputs.
- **First-Year Student Honorarium (2016):** Awarded by the Human Factors & Ergonomics Society (International).
- **James Jerger Award (2014):** For Excellence in Undergraduate Research, American Academy of Audiology.

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## CORE EXPERTISE & TECHNICAL SKILLS

### HRI & HARDWARE INTERACTION

Human-Robot/Human-AI Trust Frameworks, Multimodal Interfaces (Sound/Haptics/Light), Physical Ergonomics, Anthropometrics, Observability, Teleoperation Feedback Loops, Spatial Awareness.

### VALIDATION & RISK MANAGEMENT

Human-Systems Integration (HSI), uFMEA (Failure Modes & Effects Analysis), Formative & Summative Testing, Cognitive Load Modeling, Psychophysics, MIL-STD-1472, NASA-STD-3001.